

Diagnosing Suspects by Analyzing Human Behavior to Prevent Crime by Using Deep and Machine Learning

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Research Article

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Abstract

In both developed and developing countries, crime is undoubtedly increasing for the uncontrolled growth of the population. As a result, demands are rising to tackle issues like being concerned about everyday life and personal property through video surveillance. However, the security personnel cannot visually focus on all the CCTV footage at a time. Consequently, a system is required that we effectively recognize human behavior in real-time. Researchers have recently researched human activity recognition by machine learning and deep learning. In order to avoid fraud, we presented a procedure that uses human activity recognition to analyze a few events and determine whether or not a person is a suspect. This system labeled the video based on human action using deep learning. On the other hand, the CNN classifier is the best deep learning model for object classification, where we achieve our system with 99.98% accuracy. Furthermore, we detect suspicious based on the above classified human activity and time duration using machine learning that reaches 92.51% of accuracy. This research paper will detect suspicious through optimized, effective, and rapid. More discussion of the results, as well as limitations and future research.

1 Introduction

Human action reveals details about a person's identity, personality, and emotional functioning. Human Activity Recognition (HAR) is a critical element of suspect detection and its aim to identify activities based on inspections of subjects' activity and ambient variables [1–5]. Suspects can be classified in a variety of ways, depending on factors such as criminal behavior, crime scene trends, and motive [6–7]. Machine and deep learning algorithms can determine if a crime scene is ordered and premeditated or unorganized and unplanned [8–9]. Using a computer vision approach [10, 12], some relevant papers have been discussed about classifying objects and recognizing human action from a video [13–15] where used Long Short-Term Memory networks with convolutional neural network techniques. Another relevant paper [16, 17], who gives an overview of various classification techniques for recognizing human activities from wearable inertial sensor data. Human activity recognition or suspect detection from video surveillance [18–22] has recently been a popular topic in image processing and computer vision research [10]. Deep Learning or Machine Learning completes most of this work on HAR and suspicious detection separately [1–3]. Here, author uses a deep neural network to identify the behaviors of a lot of persons in a video frame. Due to the high-quality video process [5, 24], data travel time across the network, and the time it takes to access and present the feed on a screen, security cameras might lag or experience a delay [4, 22]. However, to mitigate such a problem, we proposed a new architecture for our system that detects suspicious activity based on video footage in real-time and provides the most efficient and reliable performance. In general, we divide our system has given architecture into two parts. The first type of data is video-based data and the second type of data is video-analyzed data.

By employing a deep learning technique in intelligent video surveillance, human activities analysis may be used for a variety of acts such as walking, running, smoking, smiling, gazing around, hitting, jumping, throwing, becoming furious, and falling and making our dataset based on video data gathered from

multiple sources such as YouTube, Stock, and social media platforms. This data can be appropriately labeled by using CNN, which has the most outstanding performance compared to other pre-trained deep learning models. Another textual dataset collects data from various social media such as YouTube, Facebook, Twitter, etc. which come from video footage labeling before the suspicious event, such as human movement and time duration for specific human activity. For the purpose of our system creates the dataset from video footage with no assumptions. The primary goal of preparing such a dataset is to provide our system with faster, more accurate, and more reliable results to prevent crime in different places. In this situation, based on this dataset use it machine learning procedure in our system predict whether or not any human activity is suspicious. Finally, the process uses a combination of deep learning and machine learning to advance our system. As a result of using this overall system to identify and analyze human activity in particular occurrences, we can defend ourselves from a variety of crimes in our environment. This study improves the system's efficiency by reducing the operational complexity of real-time suspect detection.

2. Methodology

2.1 Proposed model

Our system will be used vision-based video footage from a CCTV camera to surveillance human activity in public places to detect suspicious activity before an event occurs. Our proposed model uses two types of data which are video-based image data and video-analyzed textual data. In the Deep Learning

We found 11 different categories of activities in a video-based dataset. These are angry, fallen, hit, leap, glance around, run, grin, smoking, talking, throwing, and walk are some activities that come to connect to any crime-related situation. The full video is transformed into 56590 frames and saved as jpg files. At six activities in different positions on humans shown in fig. 2. We manually applied labels to each video data set that had the same type of activity for the same class at first.

procedure, we first collect video. Then, we can perform video and image preprocessing, pre-training model, different action classifiers, and video. We could save the title with the frame by integrating labeling with time duration where use our dataset. In machine learning approaches, we obtain after some time has passed, then examine that person's activities to see if they have committed any crimes or not, convert the absolute value into a numerical value, train with a machine learning model, and finally detect whether or not a person is suspicious.

2.2 Dataset

This paper divided the dataset into two sections:

2.2.1 Video-based image dataset

Table 1
Video dataset description

Actions	11
Video	300
Video per Action	12–17
Total Duration	2500 sec
Frame Rate	30fps
Image Size	64 X 64
Total image	56690

action in the second phase of converting video to picture. The next stage combines picture preprocessing, including histograms, normalization, and RGB to gray image conversion. We can see the regions to differentiate between images after normalizing the image histogram. The entire dataset is manually labeled and divided into three sections: 75% for training, 20% for testing, and 5% for validation. The following are the details of our created dataset:

For object classification, we use CNN model. To verify the effectiveness of the CNN in the detection of human activity, we chose a basic structure. Four Conv3D layers, two max-pooling layers, and two fully connected layers make up the model's structure. We apply 32 filters to the first pair of Conv3D layers in the default parameters and 64 filters to the second pair. The model uses an Adam

2.2.2 Video-analyzed textual data

In our dataset, we use nine factors. We divide these factors into two parts for prediction. One is independent factors, and the other one is target factors. Initially, we started with eight input factors. They are Time1, Time2, Time3, Time4, Event1, Event2, Event3, and Event4. So when we watched a video on YouTube, we divided the video into four parts Time1, Time2, Time3, and Time4. Suppose a person is walking for 10 seconds.

Table 2
Textual dataset features

Factors	Value
Time1, Time2, Time3, Time4	Continuous
Event1, Event2, Event3, Event4	Categorical Value
Target Factors	1 0

Then after completing 10 seconds, person is standing for 7 seconds. Next 5 seconds, this person eyes are working up-down, and then next 6 seconds, this person out of CCTV footage. So, the first 10s is called Time1, and the movement of the person inside 10s is called Event1. According to the action and time, we divided the dataset into Time2, Event2, Time3, Event3, Time4, and Event4.

Table 3
Textual dataset description

me1	Event1	Time2	Event2	Time3	Event3	Time4	Event4	Target
0.00-0.02	Walking	0.03-0.08	Eyes Up Down	0.09-0.17	Hit	0.18-0.29	Out of Frame	1
0.00-0.04	Out of Frame	0.05-0.10	Running	0.11-0.13	Out of Frame	0.14-0.17	Out of Frame	0
0.00-0.05	Walking	0.06-0.18	Standing	0.19-0.40	Eyes Up Down	0.41-1.10	Out of Frame	1
0.00-0.04	Running	0.05-0.13	Running	0.14-0.24	Running	0.25-0.39	Running	0
0.00-0.06	Out of Frame	0.07-0.14	Walking	0.09-0.30	Standing	0.31-0.38	Throw	1
0.00-0.04	Walking	0.05-0.10	Hit	0.11-0.14	Fall	0.15-0.18	Running	1

2.3 Convolutional neural networks as the feature extractor

optimizer and cross-entropy for loss calculation, and all kernels are 3 x 3 x 3[25]. It has two dense layers with 128 and 11 neurons at the end of the model. There are 11 different human activities represented in the output, which is multiclass.

3. Experiment And Result Analysis

The proposed model experiments and their outcomes are presented in this section. In our architecture, Activity recognition performance is evaluated for efficiency and effectiveness. The first experiment aims

to recognize and classify human activity sequences using time duration data gathered from CCTV footage. A deep learning approach was used to fulfill this section. The second experiment uses machine learning to obtain reliable estimates for the best architectures found in the first experiment, which can be used for further testing in various situations. To evaluate our model's performance, we created our dataset with the goal of HAR and suspicious detection.

3.1 Human activity recognition with timing use deep learning:

3.1.1 Test data performance of activity classification use confusion matrix

As an evaluation technique, all methods' confusion matrices and classification reports were compared. A confusion matrix is a table used to define a classification algorithm's effectiveness. The effectiveness of a classification algorithm is envisioned and interpreted using a confusion matrix. Table 4 shows

Table 4. CNN model confusion matrix

Angry	1412 100%	0	0	0	0	0	0	0	0	0	0	1412 100%
Fall	0	647 99.8%	0	1 0.2%	0	0	0	0	0	0	0	647 99.8%
Hit	0	0	768 100%	0	0	0	0	0	0	0	0	768 100%
Jumping	0	0	0	1831 100%	0	0	0	0	0	0	0	1831 100%
Looking around	0	0	0	0	1788 100%	0	0	0	0	0	0	1788 100%
Run	0	0	0	0	0	516 100%	0	0	0	0	0	516 100%
Smile	0	0	0	0	0	0	1782 100%	0	0	0	0	1782 100%
Smoke	0	0	0	0	0	0	0	990 100%	0	0	0	990 100%
Talk	0	0	0	0	0	0	0	0	810 100%	0	0	810 100%
Throw	0	0	0	0	0	0	0	0	0	1464 100%	0	1464 100%
Walk	0	0	0	12 0.6%	0	0	0	0	0	0	1944 99.4%	1944 99.4%
	Angry	Fall	Hit	Jumping	Looking around	Run	Smile	Smoke	Talk	Throw	Walk	Precision recall accuracy

the confusion matrix and classification report of the CNN classifier. Each lab-el represents each class of activity: anger, fall, hit, jumping, looking around, running, smiling, smoking, talking, throwing, and walking. We are divided video-base dataset where 20% data, which means 13965 images belonging to 11 classes use test data in the CNN model. Our dataset has been able to separate each category using the model very effectively. In addition, we are experimenting with other deep learning techniques, which are nearly identical, but the convolution neural network performs best. As a result, we have decided on a more realistic feature extraction effect.

3.1.2 CNN model training accuracy and loss

The entire dataset is manually labeled and divided into three sections for training, testing, and validation. Using CNN as the pre-trained model to train our training dataset with 30 epochs and a batch size of 64, we achieved a Test Loss: 0.005 and Train Loss: 0.004 and Test Accuracy: 99.98%, and Train Accuracy: 100%.

We can see from the Figs. 5 and 6 graphs that the model's accuracy has become more linear after 90% accuracy, indicating that accuracy is between 90% to can conclude that it has also reached a point where it will vary substantially.

3.1.3 Comparison between different deep learning models accuracy

CNN had the best result in this study's classical deep learning category, with 99.9% accuracy. A convolution neural network classification model is used to improve the predictability of physical activity. A score of 99.9% accuracy has been achieved. IN Fig. 6 compared to accuracy performance measures, precision, recall, and F1 score provide more insight into the prognostication. The F1 score offers the same data as the recall value. The precision value shows in Fig. 6 that the F1 score and recall are nearly identical.

Best classification result in CNN's model accuracy score after 30 epochs achieve around 99.9%, that the highest among the pre-trained models. After 30 epochs, VGG-16 achieves 92% accuracy, which is comparable to the CNN model. When comparing the accuracy of other models to ResNet50, there was considerable discrepancy. After comparing all of the models, we can conclude that VGG-16 and CNN were the most stable and accurate.

3.1.4 Proposed method feature extractor result

We previously stated that, among these three models, we have analyzed and compared the results to determine that the CNN model is the best fit for us. Check out some human activity videos to see how well our deep learning approach detects it. CNN is used for frame-level feature extraction. Our system also detected human action with timing at the same time. When displaying the output, choose a random human position to time. After reading a frame from the input video, it decided if the structure contained which action human at present. This method uses a larger number of vision-based datasets to detect suspicious behavior using machine learning. Any suspicious real-world behavior video is fed into our model, determining how accurately the video is labeled. We are now using this model to create a machine learning dataset with text data.

3.2 Suspect detect use machine learning approaches:

3.2.1 Classification report

Precision, Recall, and F1 score should ideally be 1 (high) for a good classifier. From Table 5, we see that in the Precision column, there is no 1 score, but values have close to 1. So, based on Precision, SVM, Random Forest, and Logistic Regression are the best algorithms among the above 5.

From Table 5, we see that there is two different model achieve 1 Recall value which are well good in this system. So, based on the Recall, Random Forest and Logistic Regression are the best algorithms among the above five algorithms. Only when precision and recall are both ones do the F1 score become 1. Only when both precision and recall are high does the F1 score rise. The F1 score is better than accuracy because it is the harmonic mean of precision and recall. From this table, we also see that in the F1 Score column, there is no 1, but values have close to 1. So, based on the F1 Score Random Forest and Logistic Regression are the best algorithms among the above five algorithms.

Table 5
Classification report for all algorithms

Algorithm	Precision	Recall	F1 Score
Random Forest	0.87	1.00	0.93
Decision Tree	0.86	0.92	0.89
SVM	0.88	0.88	0.88
Logistic Regression	0.87	1.00	0.93

3.2.2 Classification results

Finally, we applied machine learning techniques called Decision Tree, Random Forest, K-Nearest Neighbor, Support Vector Machine, and Logistic Regression. The number of correctly classified data instances divided by the total number of data instances is accurate. We can see the testing accuracy in Random Forest is 0.92, and the training accuracy is 0.95. Then the testing accuracy in Logistic Regression is 0.92, and the training accuracy is 0.94. Of all the algorithms, those two algorithms' accuracy is best fit for our dataset. The comparison between different algorithm's accuracy is shown in the bar chart in Fig. 9 below:

3.2.3 ROC curve of marching learning model

A ROC is a graphical plot used as a performance measurement for classification problems. The true positive rate (TPR) is mapped against the false Positive rate (FPR) to generate the ROC curve. Over the total number of data instances, accuracy represents the number of correctly classified data instances. A possibility slope plots of the TPR against the FPR have various criteria, effectively, and greatest performance was shown by the topmost point of the curve.

3.2.4 Feature importance of Our textual Dataset

Techniques that assign a score to input features based on how useful they are at predicting a target variable are known as feature importance. It refers to methods for calculating a score for each model's

input features; the scores represent the "importance" of each segment. A higher score indicates that a particular component will impact the model used to predict a specific variable. It is one thing to build a model, but it is quite another to comprehend the data that goes into it. Like a correlation matrix, feature importance helps to us understand the relationship between the features and the target variable. It also aids us in determining which parts are unimportant to the model.

From Fig. 11, we see that Time4end provides the highest score means it will have a more significant effect on the model. On the other hand, Time1Start provides the lowest score, not affecting the model.

3.3 Compared to other available datasets and performs

In recent times, attempts have been made to solve many problems by analyzing the use of human beings, and many people are currently researching the subject. The three data sets presented above were compared and contrasted in various ways. We compare the number of data points per activity in this data set in Fig. 12. The number of times each activity is conducted in the UCI HAR data set [26–28], WISDM data set [26, 29, 30], and UCF101 data set [31–33] varies significantly. The UCF101 data has 101 action types, while the other datasets have six classes (activities), which differ in activity and size. Both datasets were labeled by hand as well. Furthermore, this data set applies a deep learning approach, and the inaccuracy of the results is shown in the graph. All of the datasets are based on thesis work on human behavior. Finally, the purpose of a human activity recognition system is our own more active and accurate dataset than this data.

Table 6
Accuracy comparison other work

Previous Work	Present Work
RestNet18 = 0.90	Random Forest = 0.92
RestNet34 = 0.91	Logistic Regression = 0.92
RestNet50 = 0.93	KNN = 0.89

On the other hand, once the HAR first process is complete, we move on to the second process, which uses machine learning to determine whether or not something is suspicious. This section will compare the current paper [10], which detects suspicious activity based on five suspicious activities. Shooting, punching, kicking, knife attack, and sword fighting are among the activities. RestNet18, RestNet34, and RestNet50 were among the algorithms used. On the other hand, we went about it differently. Our model predicts whether a person is suspicious or not based on their movements. We used a machine learning approach and algorithms such as Decision Tree, Random Forest, and Support Vector Machine. The previous work accuracy is between 0.90 and 0.93, according to Table 6. Their deep learning algorithm has the highest accuracy of 0.93 and the lowest accuracy of 0.90. On the other hand, considering the top three machine learning algorithms in this study, the highest accuracy is 0.92, and the lowest accuracy is 0.89.

4. Conclusion And Future Work

Before any incident occurs, some suspicious movement or behavior of a person can be observed. Any analysis of at all given time or two or one activity based on the use of the last moment does not imply that any crime will occur in the future.

This paper investigates the object and human activity recognition and behavior analysis model from CCTV footage using a combination of deep learning and machine learning where use our dataset. First, recognize human activity; then, after some time has passed, analyze this person use to determine whether or not that person has committed any crimes. This method can lead to increased caution before an incident occurs when this method is used. On the other hand, when we detect suspicious through use analysis, we can analyze more textual data in a shorter period, thus reducing the complexity of the method compared to the direct image or video analysis. So, our system eliminates heavy computational methods, and the real-time automatic suspicious detection System seems to be accurate in less time will give feedback.

To control all security systems purpose, we plan to increase the number of activities, including complex physical activities, and handle suspicious related situations for the experiment. More camera types and analyses will be added, allowing us to expand our data sets in other dimensions. This system analyzes human behavior for a few events and times, but the results are unreliable; thus, the next challenge will be to monitor human behavior for more extended periods to discover suspicious behavior.

Declarations

Ethics declarations

Conflict of interest

The authors declare that they have no conflict of interest.

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Not applicable.

Availability of Data and Material

Because of the size and propriety of the dataset used for this research, it could not be included in this manuscript.

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Figures

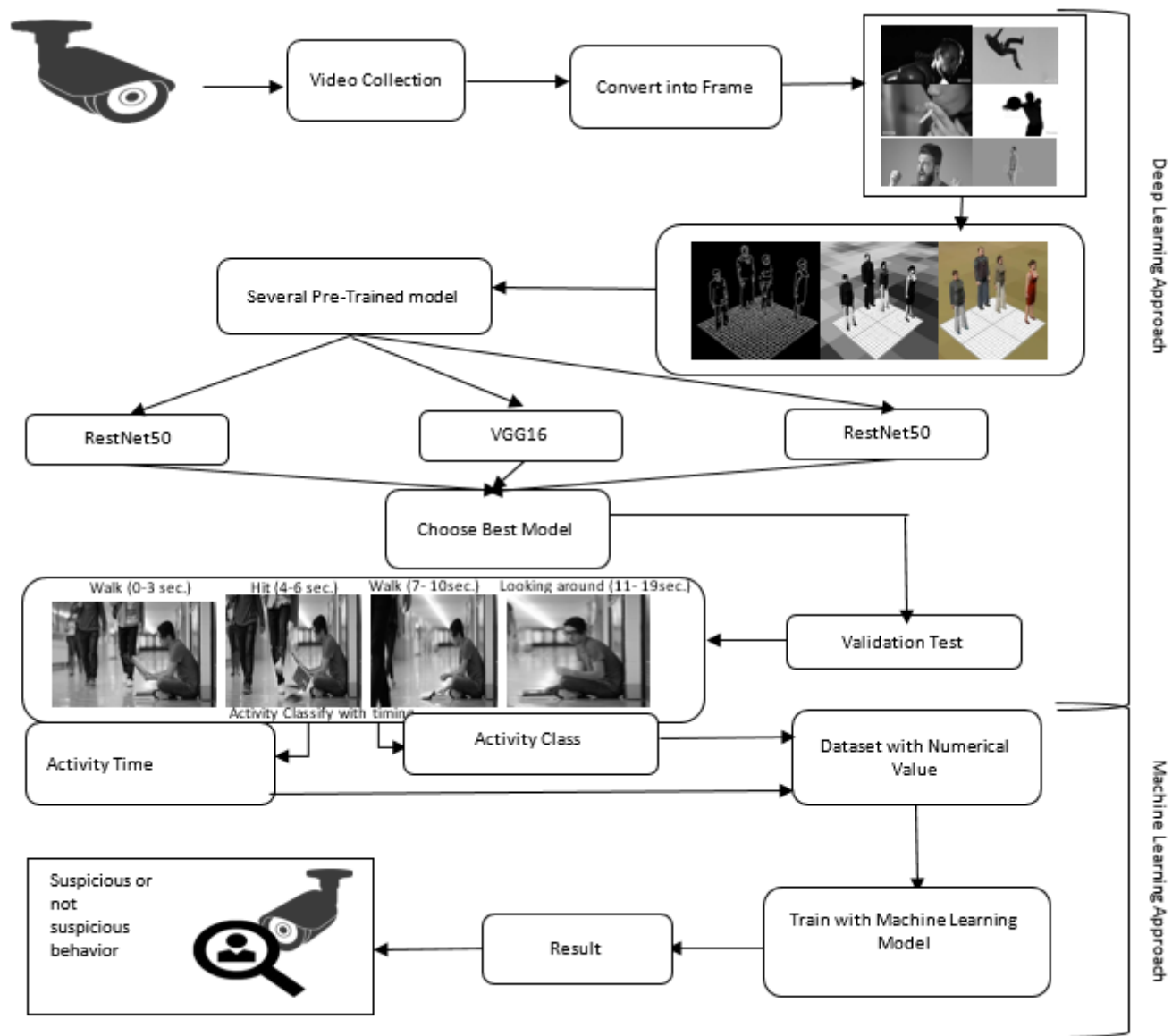


Figure 1

System architecture

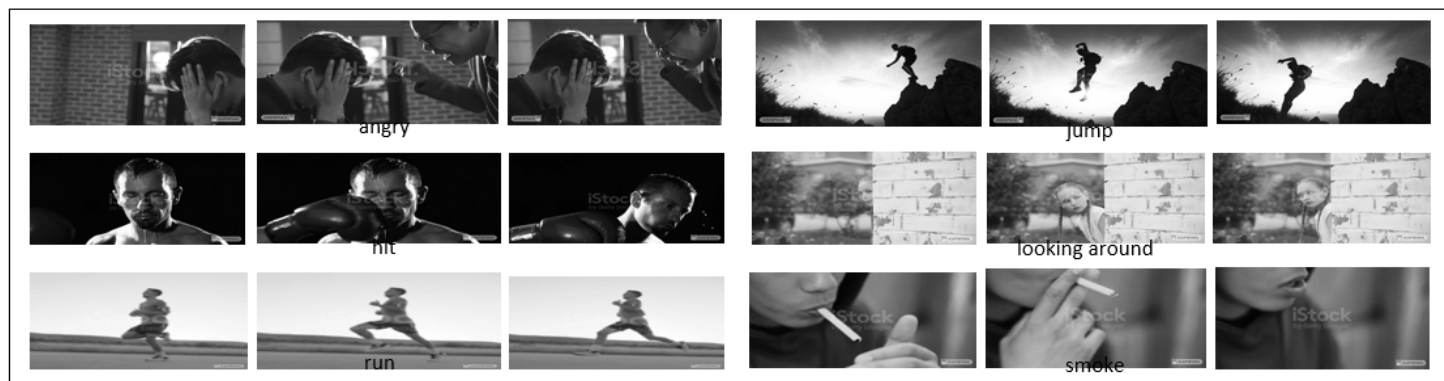


Figure 2

Video dataset sample

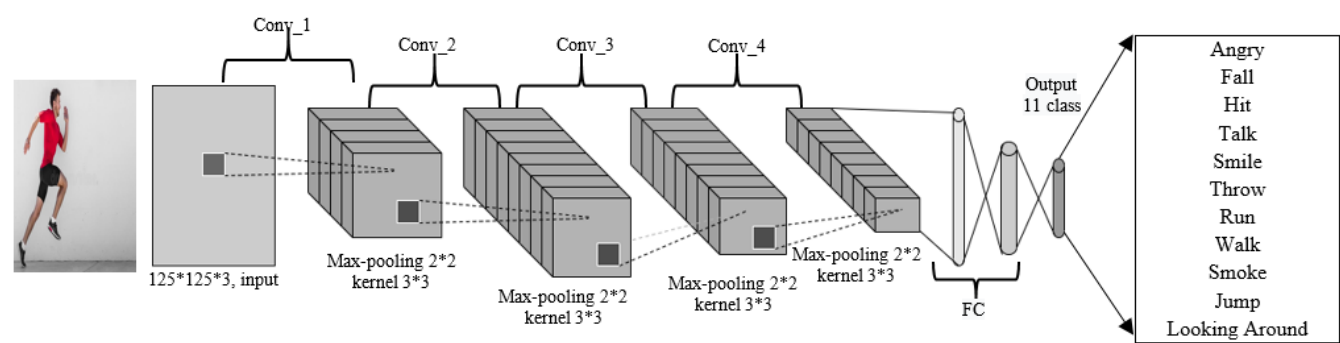


Figure 3

The architecture of the CNN Mode

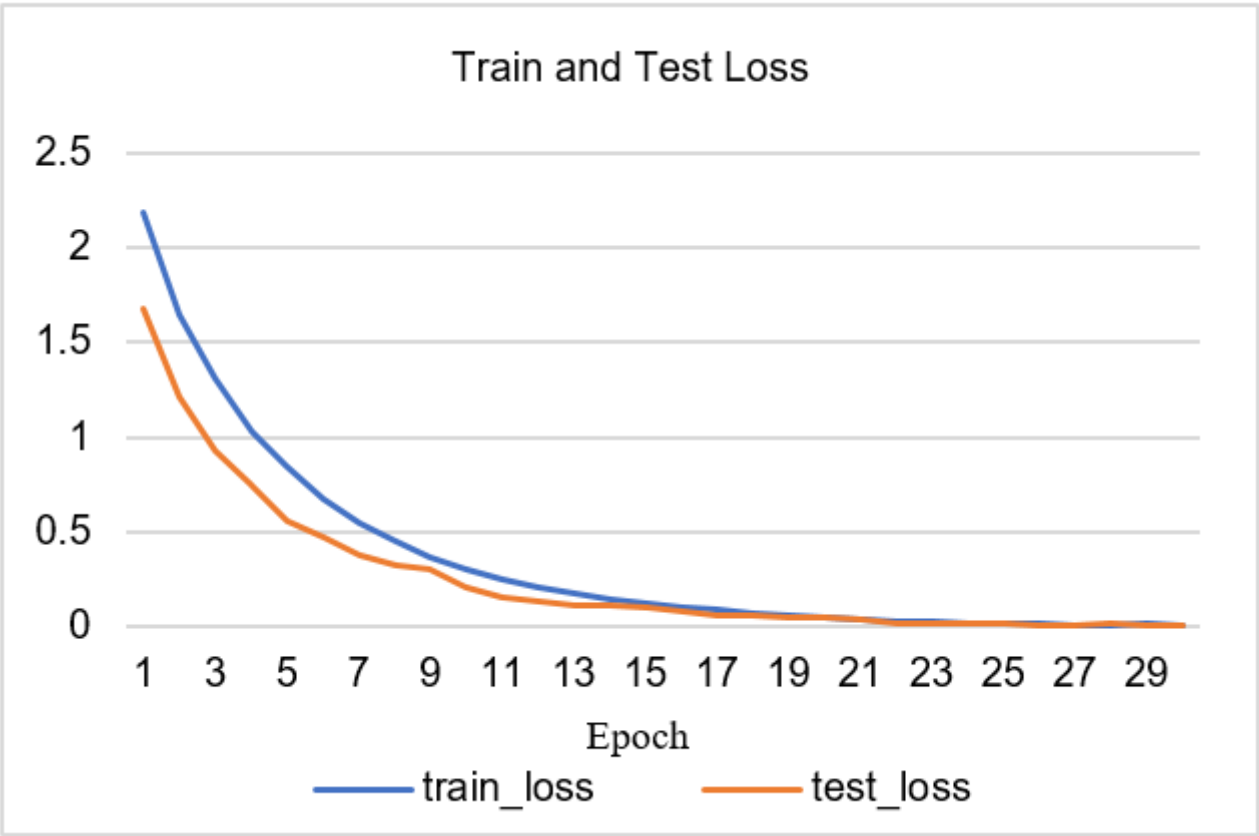


Figure 4

CNN model train and test loss data

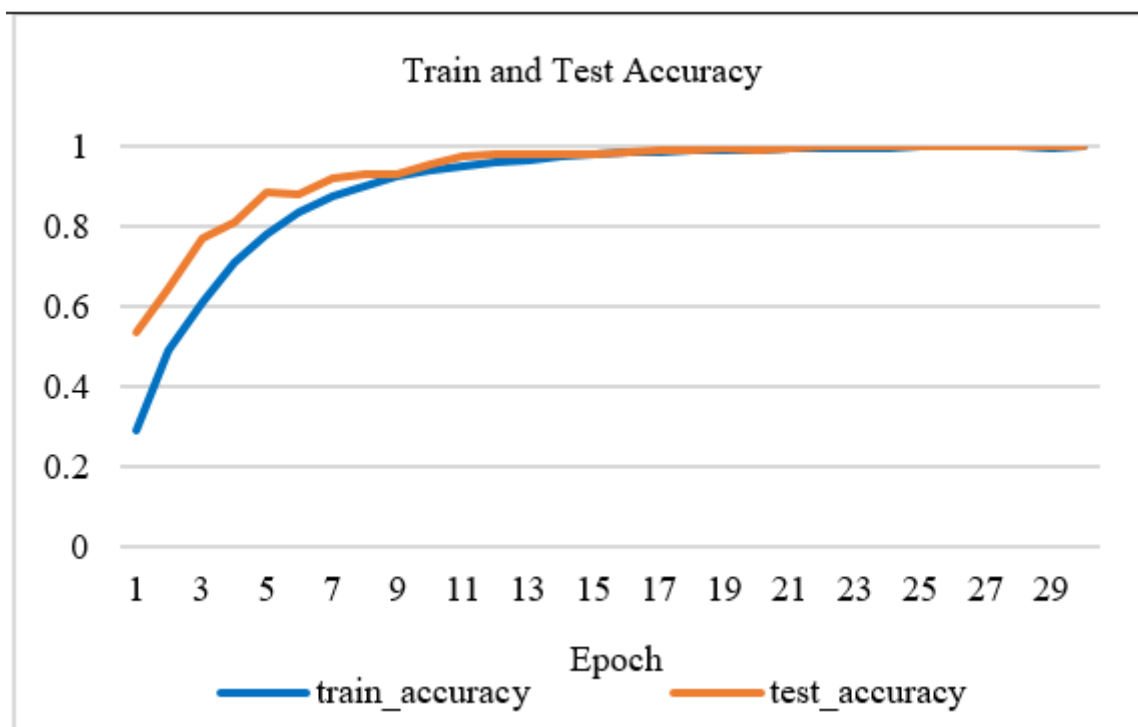


Figure 5

CNN model train and test accuracy data

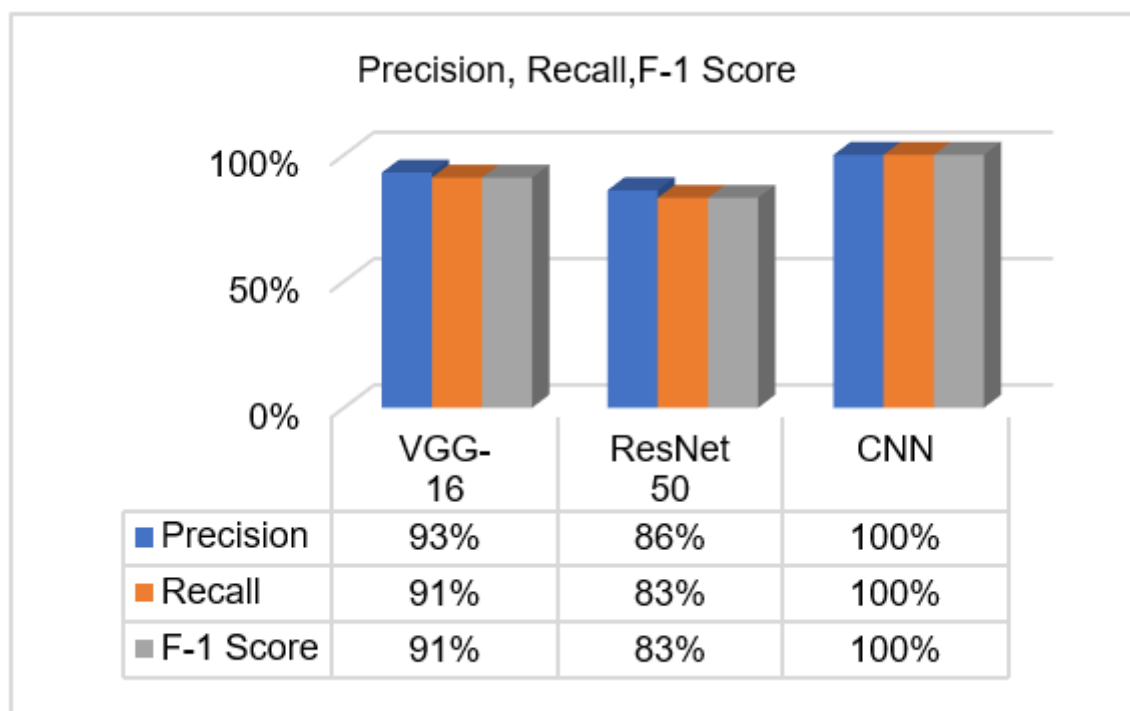


Figure 6

Different model Precision, Recall, F-1 Score

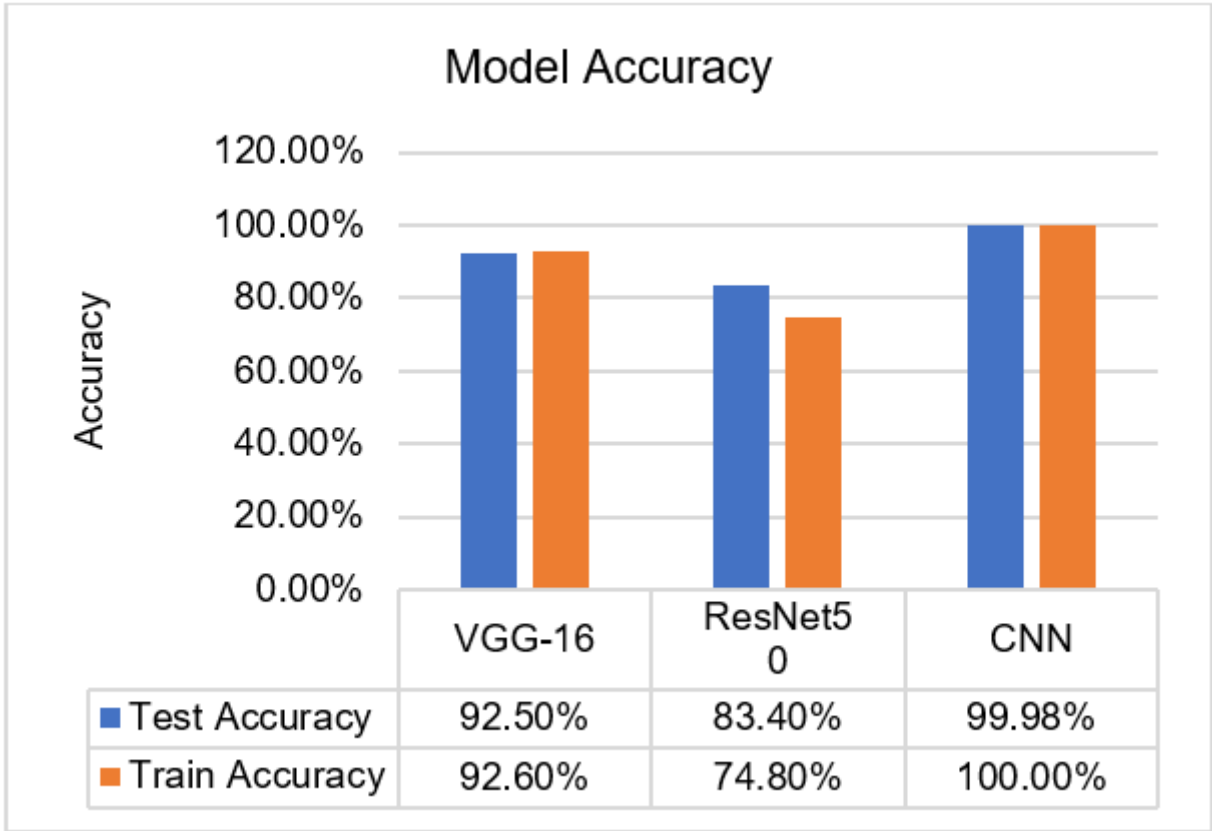


Figure 7

Deep learning model accuracy

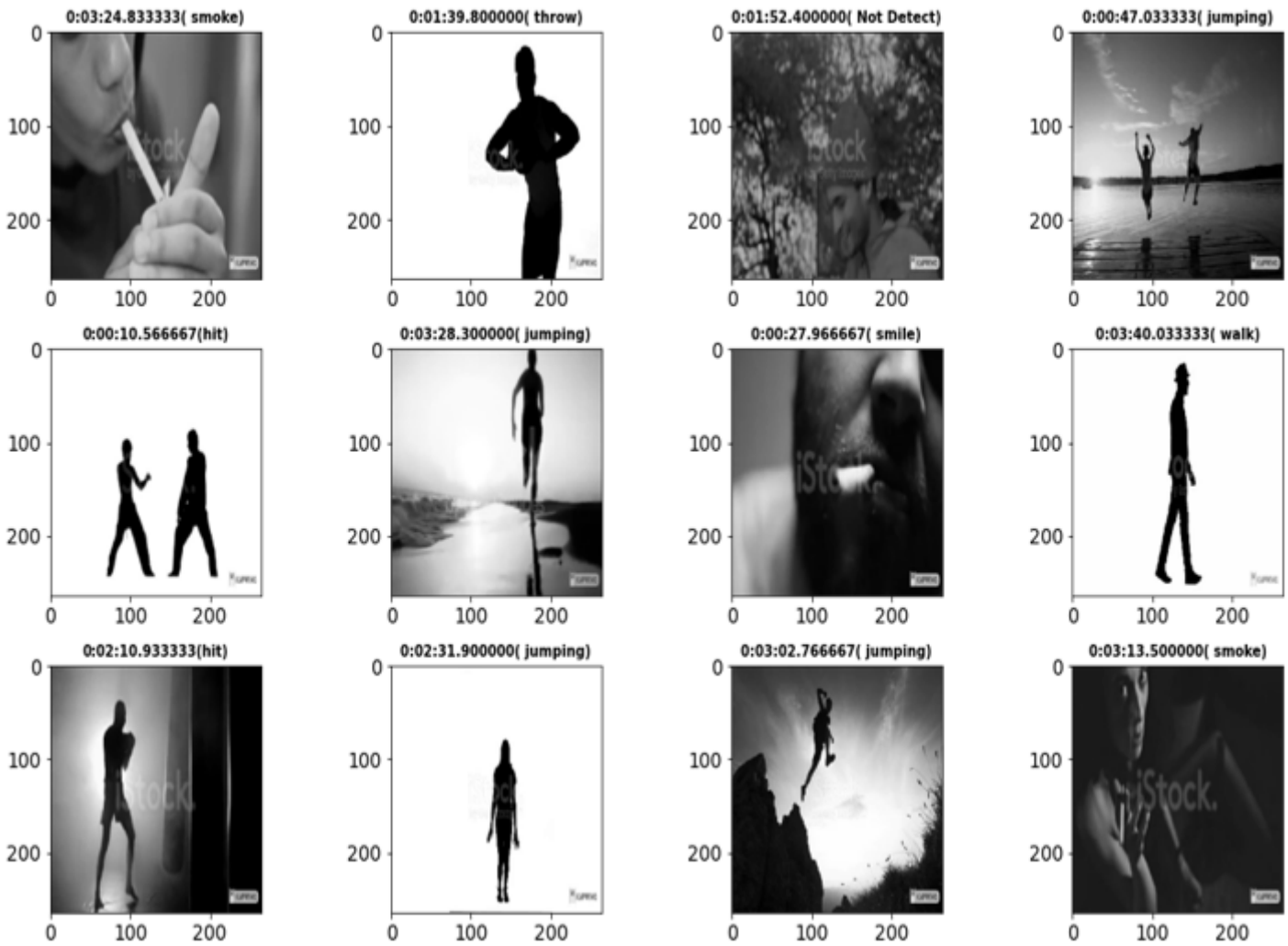


Figure 8

Feature Extractor Test Output

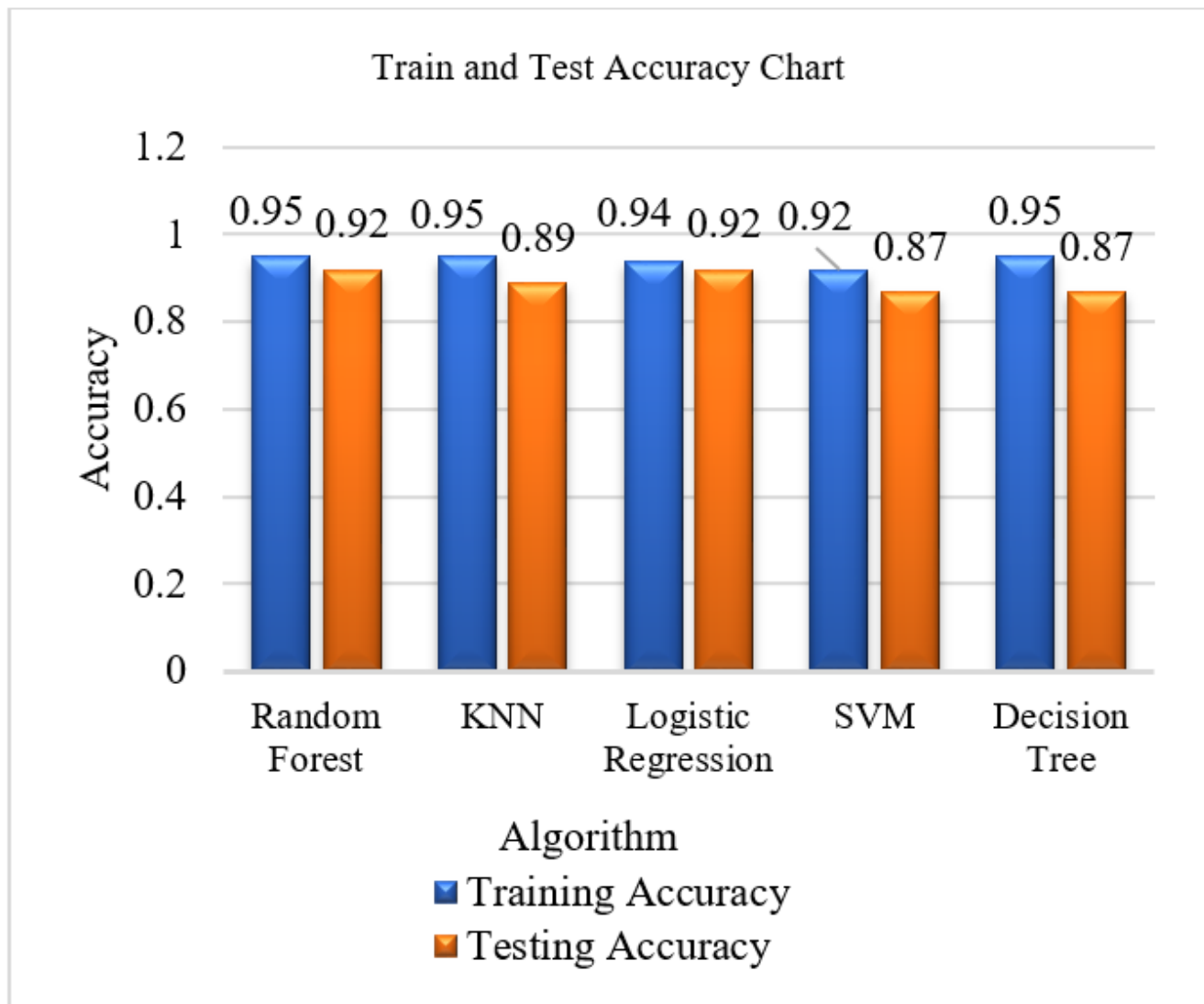


Figure 9

Accuracy comparison report visualization

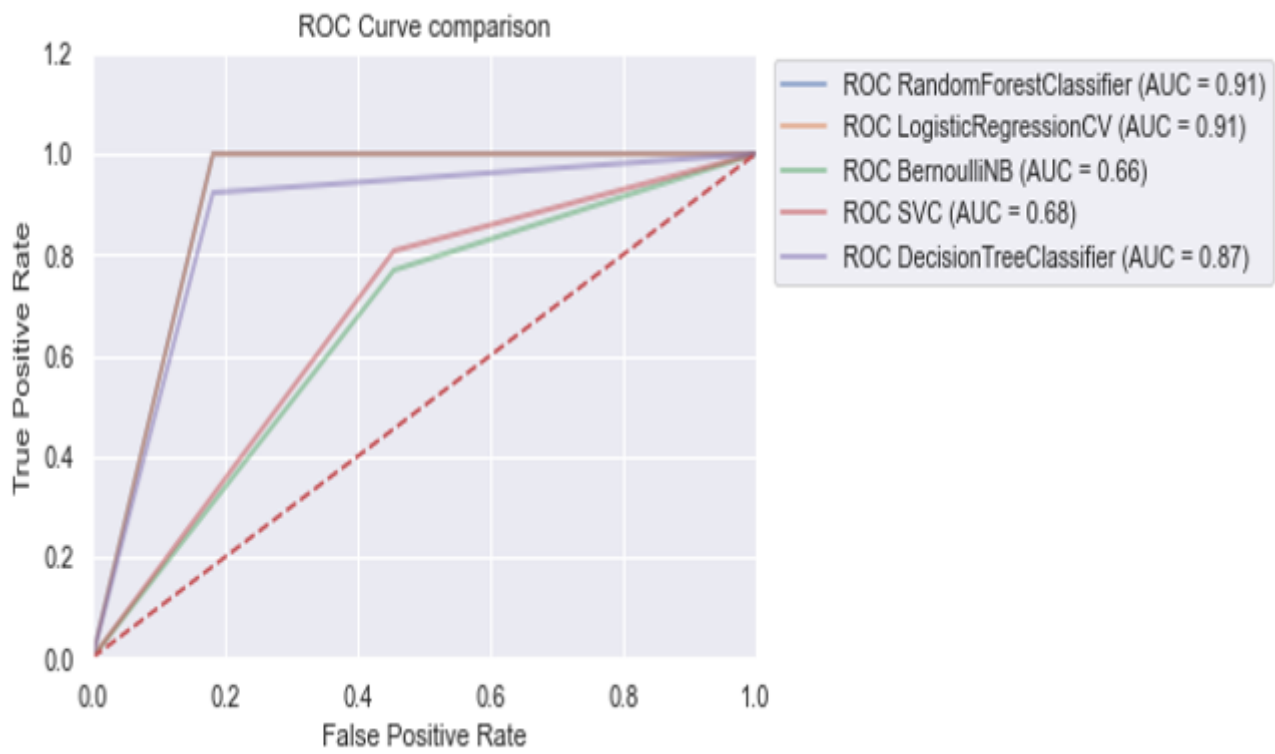


Figure 10

ROC curve comparison between different algorithms

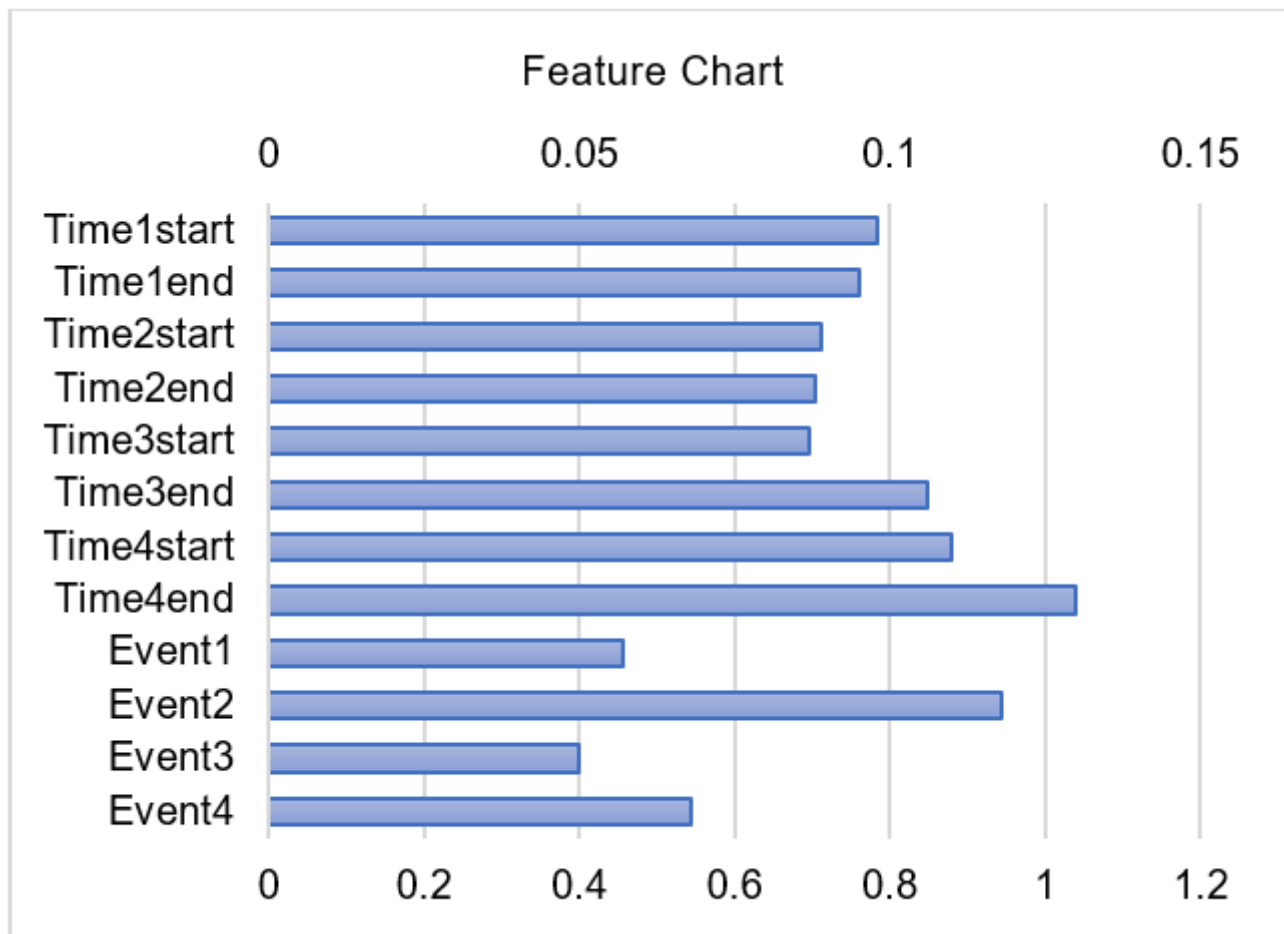


Figure 11

Feature importance

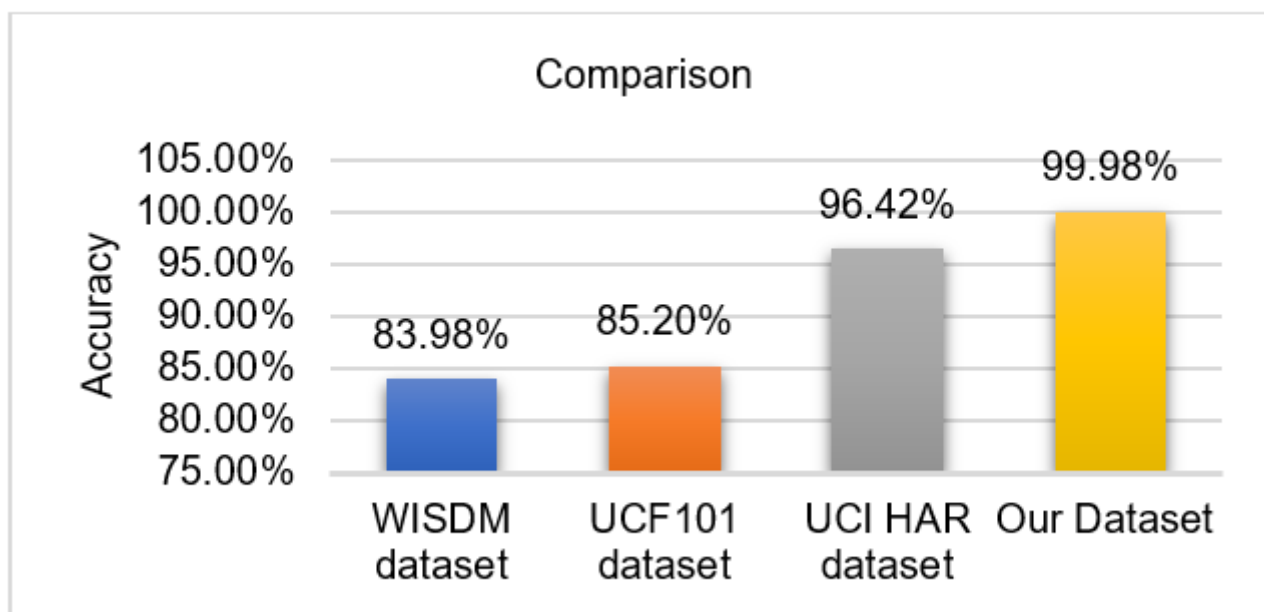


Figure 12

Comparison between different dataset and accuracy

